

The Way to Go – Extended Study Guide for Backend Development

This guide provides a complete breakdown of what to study, what to skip, and how to structure your weekly learning. It includes detailed explanations, a backend learning roadmap, and diagrams of dependency order. Designed for your specific goal: learning Go fundamentals → REST APIs → authentication → multi-device sessions → microservices.

1. Detailed Table of What to Learn / Skip

Chapter	Subtopic	Status	Explanation
2	Basic Types, Variables, Strings, Pointers	Learn	Required to understand structs, requests, responses, and memory management.
2	Bitwise Ops, Rune Details	Skip	Not needed for backend engineering.
3	If/Switch/For	Learn	Critical for handler logic and validations.
4	Functions, Defer, Error Patterns	Learn	Key for request lifecycle and database cleanup.
4	Recursion/Advanced Closures	Skip	Not useful for API development.
5	Slices & Arrays (Basics)	Learn	Slices appear everywhere in backend code.
5	Multidimensional Arrays	Skip	Rarely needed.
6	Maps	Learn	Useful for configs, JSON parsing, caching.
7	Packages, go test, External Packages	Learn	Essential for structuring real backend applications.
7	regex, sync (Advanced)	Later	Important only for advanced systems.
8	Structs, Methods, Receivers, Interfaces	Learn	Core of any backend or microservice design.
8	Reflection, Empty Interface	Skip	Avoid until advanced.
10	JSON Encode/Decode	Learn	Mandatory for API communication.
10	XML, gob, Wiki Reader	Skip	Not used in modern backend flows.
11	Error Handling, Panic/Recover, Testing	Learn	Crucial for stability.
12	Goroutines Basics, Channels Basics	Later	Useful after mastering HTTP.
12	Advanced Concurrency Patterns	Skip	Only for high-performance systems.
13	Simple Web Server	Learn	Foundation for REST APIs.
13	TCP Low-level Networking	Skip	HTTP frameworks handle this.
14	Pitfalls & Idioms	Learn	Helps avoid subtle bugs.
15	Performance Tuning	Later	Learn after shipping a real service.

2. Weekly Study Plan

Week 1: - Finish Chapters 2–3 essentials - Practice small programs: slices, maps, structs
Week 2: - Finish main parts of Chapter 4–5–6 - Build a mini Go module - Write small JSON marshal/unmarshal programs
Week 3: - Start HTTP server with net/http - Switch to Gin - Build /health and /photos endpoints
Week 4: - Build auth: register/login (in-memory) - Add bcrypt + JWT - Add sessions table (database)

3. Dependency Order (Text Diagram)

Go Basics → Structs → JSON → HTTP → Routing → Auth → Sessions → Token Logic → Database → Microservices → Concurrency

4. Backend Learning Roadmap

1. Go basics (variables, types, structs, methods) 2. JSON encode/decode 3. net/http → Gin framework 4. Routing, middleware, handlers 5. Database layer (Postgres) 6. Authentication and authorization 7. Session management (multi-device login) 8. Docker and docker-compose 9. Microservice separation 10. Optional: concurrency patterns, caching, rate limiting

5. Notes Section

(Use this page to write your own notes during study)